Python File Handling and CSV Complete Revision Notes

Definitions

open(): Used to open a file.

read(): Reads the entire file and returns one string.

readline(): Reads one line at a time.

readlines(): Reads all lines and returns a list.

close(): Closes the file and releases resources.

r mode: Read mode. File must exist.

w mode: Write mode. Creates file or overwrites existing content.

a mode: Append mode. Adds content at the end of file.

csv.reader(): Reads CSV data row by row and returns lists.

next(reader): Skips the header row.

row[0], row[1]: Access individual columns from a row.

csv.writer(): Used to write rows to a CSV file using lists.

writerow(): Writes a single row.

writerows(): Writes multiple rows.

csv.DictReader(): Reads CSV rows as dictionaries.

csv.DictWriter(): Writes dictionary data to a CSV file.

fieldnames: Column names used by DictWriter.

writeheader(): Writes the column headings/header row.

newline="": Prevents extra blank lines in CSV files.

Mistakes Encountered and Lessons Learned

• Wrong folder path in Terminal.

• Used system.argv instead of sys.argv.

• Unterminated string literal in open().

• Confusion about read(), readline(), readlines().

• Expected file update without understanding file closing.

• Mixed tabs and spaces causing TabError.

• Confusion about csv.reader output [HELLO GUYS].

• Used fieldnames with csv.writer instead of csv.DictWriter.

Quick Differences

• read() -> Complete file as string

• readline() -> One line

• readlines() -> List of lines

• reader() -> Lists

- DictReader() -> Dictionaries
- writer() -> Lists
- DictWriter() -> Dictionaries